

EN401 — Handout 4 Results & Analysis Report

Upstream Fuel Supply Options & Economic Merit Order Behavior

1. Scenario Objective & LP Solution Metrics

- **Objective Value (NPV):** \$51,009,521.16 (0.0% change vs HO3)
- **Problem Scale:** 872 constraints, 480 variables, 4,320 non-zeros
- **Technologies Added:** MINNGS, IMPDSL

2. System Cost Invariance Analysis

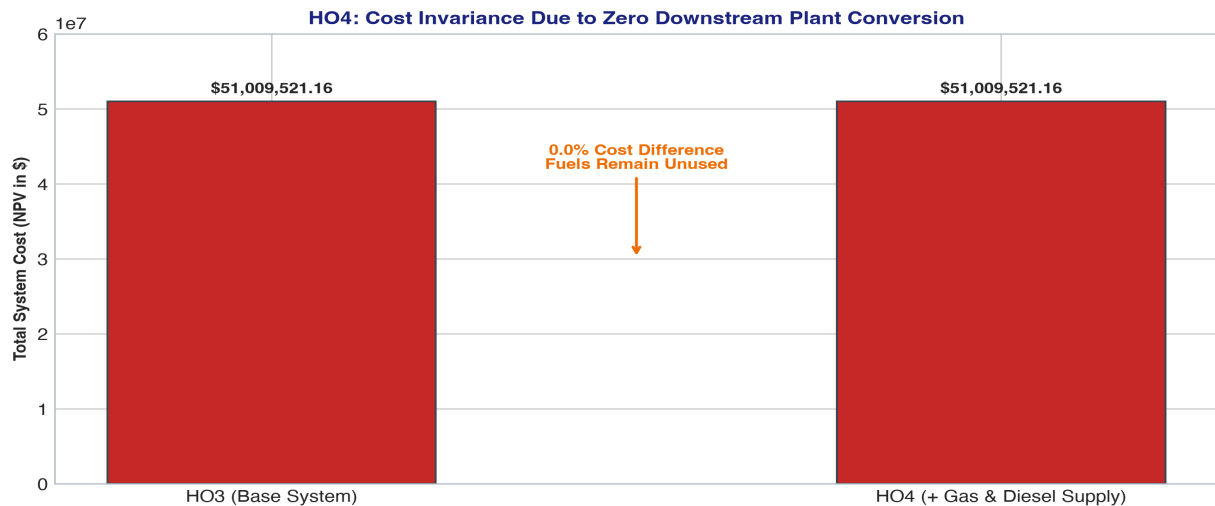


Figure: Total discounted cost remains unchanged between HO3 and HO4.

3. Engineering & Modelling Takeaways

The solver strictly adheres to network connectivity. Introducing primary fuels without power generation assets leaves the fuels stranded. BACKSTOP continues to cover demand until power plants are introduced in HO5.